

A Mechanically Verified Translation of $MTL_{0,\infty}$ into Timed Büchi Automata

Elie Fares, Jean-Paul Bodeveix, Hussain Al-Aqrabi, and Azar Salami



Abstract—We present a Coq-verified translation from pointwise dense-time $MTL_{0,\infty}$ to timed Büchi automata. The translation first reduces ordinary timed operators to strict hatted primitives, encodes their timing constraints in clocked LTL, and then uses an arbitrary LTL-to-Büchi backend satisfying an explicit semantic contract. A relaxation and reset completion construction turns the backend automaton into a timed Büchi automaton. The central proof difficulty is reuse: one clock is assigned to each distinct primitive timed subformula, even when its obligation is activated repeatedly and overlaps itself. We establish soundness and completeness by a canonical reset policy and residual obligations, and prove the end-to-end theorem `MTL_to_TBA_correct_with`. The result preserves strict and non-strict bounds and does not depend on a particular LTL-to-Büchi implementation.

Index Terms—Metric temporal logic, timed Büchi automata, Coq, formal verification, clocked LTL, semantic preservation.

1 INTRODUCTION

Real-time requirements combine qualitative ordering with quantitative timing. Metric Temporal Logic (MTL) expresses both, while timed Büchi automata (TBA) provide a finite-state model suitable for timed verification. A translation between them is useful only when the automaton preserves the pointwise dense-time semantics of the formula. The translation itself is subtle: a timed subformula nested under an unbounded temporal operator may be activated without bound, and several instances of the same obligation may be pending at once.

This paper gives a mechanized translation for the one-sided-bounded fragment $MTL_{0,\infty}$ [1]–[3]. Its key construction assigns a clock to each *distinct primitive timed subformula*, rather than to each syntax-tree occurrence or run-time activation. The clocked-LTL encoding constrains that shared clock through reset and preservation markers. The proof’s completeness direction chooses a canonical reset trace that retains the oldest useful reference when it subsumes newer obligations; residual predicates express what remains of an obligation after time has advanced. This establishes that repeated activations need no additional clocks.

The translation is factored through ordinary LTL. It is therefore independent of the particular LTL-to-Büchi translator, subject only to a stated correctness contract on propo-

sitional words. After the backend, relaxation removes irrelevant negative extended-atom literals, and reset completion turns the relaxed cubes into timed transitions. The resulting theorem is mechanized in Coq for all eight hatted timed primitives, including strict bounds. The proof does not cover parsing, the interface to an external LTL translator, or output printing; these are outside the theorem.

Our contributions are:

- a pointwise-semantics-preserving clocked-LTL encoding of the eight hatted Until and Release primitives, with ordinary timed operators derived by semantic identities;
- soundness and completeness of that encoding, including a canonical reset policy and residual obligations that justify one clock per distinct timed subformula despite arbitrarily overlapping activations;
- relaxation and reset completion, with an end-to-end Coq theorem for every LTL-to-Büchi backend satisfying the explicit correctness contract.

The formal development includes the named results `EncodingCorrect_proved`, `T_uses_every_clock`, and `MTL_to_TBA_correct_with`. The complete arguments for the canonical policy and its strict-bound cases appear in the supplementary material. This paper treats the translation through TBA construction; generic TBA optimization and UPPAAL export are separate contributions and are not needed for the theorem proved here.

2 LOGIC AND SEMANTIC MODEL

2.1 Pointwise $MTL_{0,\infty}$

Let Σ be a finite set of actions. Formulas are in negation normal form and use Boolean operators, next, untimed Until and Release, and timed Until and Release whose intervals have one endpoint at zero or infinity. We write a bound as $\sim d$, with $\sim \in \{<, \leq, >, \geq\}$. The logic also contains the hatted operators $\hat{U}_{\sim d}$ and $\hat{R}_{\sim d}$. In a hatted operator the current position is skipped by the logical interval, while its timestamp remains the timing reference.

A timed word is $w = (a_i, t_i)_{i \in \mathbb{N}}$, where $a_i \in \Sigma$, $0 \leq t_0 < t_1 < \dots$, and timestamps diverge. We write $\text{elap}_w(i, j) = t_j - t_i$. For Until, $w, i \models_{\text{MTL}} p \hat{U}_{\sim d} q$ holds when some $j \geq i$ satisfies q and $t_j - t_i \sim d$, with p at each position in $[i, j)$. For hatted Until, the witness is $j > i$ and p is required on (i, j) .

Elie Fares, Hussain Al-Aqrabi, and Azar Salami are with the Higher Colleges of Technology, Sharjah, United Arab Emirates (e-mail: efares@hct.ac.ae; halaqrabi@hct.ac.ae; asalami@hct.ac.ae).

Jean-Paul Bodeveix is with IRIT, Université de Toulouse, France (e-mail: bodeveix@irit.fr).

Release is the dual universal condition: at every position in the bounded region, either q holds or p held earlier in the corresponding interval. Hatted Release excludes the current position from that region. We use the standard pointwise semantics [4].

For precision, the four bounded clauses used below are:

$$\begin{aligned}
w, i \models_{\text{MTL}} p \mathbf{U}_{\sim d} q &\iff \exists j \geq i. w, j \models_{\text{MTL}} q \\
&\quad \wedge t_j - t_i \sim d \\
&\quad \wedge \forall k \in [i, j). w, k \models_{\text{MTL}} p, \\
w, i \models_{\text{MTL}} p \widehat{\mathbf{U}}_{\sim d} q &\iff \exists j > i. w, j \models_{\text{MTL}} q \\
&\quad \wedge t_j - t_i \sim d \\
&\quad \wedge \forall k \in (i, j). w, k \models_{\text{MTL}} p, \\
w, i \models_{\text{MTL}} p \mathbf{R}_{\sim d} q &\iff \forall j \geq i. \\
&\quad \left(t_j - t_i \sim d \Rightarrow \right. \\
&\quad \left. \left(w, j \models_{\text{MTL}} q \right. \right. \\
&\quad \left. \left. \vee \exists k \in [i, j). w, k \models_{\text{MTL}} p \right) \right), \\
w, i \models_{\text{MTL}} p \widehat{\mathbf{R}}_{\sim d} q &\iff \forall j > i. \\
&\quad \left(t_j - t_i \sim d \Rightarrow \right. \\
&\quad \left(w, j \models_{\text{MTL}} q \right. \\
&\quad \left. \vee \exists k \in (i, j). w, k \models_{\text{MTL}} p \right).
\end{aligned}$$

The index intervals refer to event positions, whereas $t_j - t_i$ is physical time. This distinction is material when no event occurs exactly at a bound. The strict increase assumption excludes simultaneous events; divergence is used in the lower-bounded Until fairness argument.

The well-formedness condition requires $d > 0$ for every lower bound and strict upper bound, and $d \geq 0$ for a non-strict upper bound. Ordinary timed operators are definitions in the mechanized core. For upper bounds $\lesssim \in \{\leq, <\}$ and lower bounds $\gtrsim \in \{\geq, >\}$, they are:

$$p \mathbf{U}_{\lesssim d} q \equiv q \vee (p \wedge (p \widehat{\mathbf{U}}_{\lesssim d} q)), \quad (1)$$

$$p \mathbf{U}_{\gtrsim d} q \equiv p \wedge (p \widehat{\mathbf{U}}_{\gtrsim d} q), \quad (2)$$

$$p \mathbf{R}_{\lesssim d} q \equiv q \wedge (p \vee (p \widehat{\mathbf{R}}_{\lesssim d} q)), \quad (3)$$

$$p \mathbf{R}_{\gtrsim d} q \equiv p \vee (p \widehat{\mathbf{R}}_{\gtrsim d} q). \quad (4)$$

These definitions separate the current-position case from the strictly future obligation. Their pointwise semantic compatibility, including the stated bound hypotheses, is proved in Coq:

Theorem 2.1 (Semantic compatibility of the derived timed operators). *Under the well-formedness conditions above, Eqs. (1)–(4) agree with the standard pointwise semantics for all strict and non-strict upper and lower bounds.*

The identities follow by separating the current-position case from a strictly future endpoint. For upper Until, a witness at i contributes q ; every later witness requires p at i and leaves the hatted obligation. A lower bound with $d > 0$ excludes i , so p is required now and the remaining witness is hatted. The Release identities are the dual case split: an upper-bound region includes i and therefore requires q there, while a lower-bound region excludes i . Strict upper bounds also require $d > 0$, avoiding the degenerate < 0

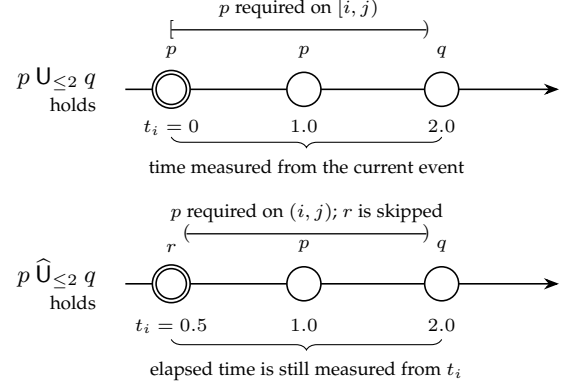


Fig. 1. Ordinary and hatted Until. The double circle is the timing reference. The hatted operator skips the current event logically but measures from its timestamp.

case. The eight corresponding Coq lemmas are listed in Supplement 2.2. Thus only the eight hatted timed constructors require primitive translation clauses.

This distinction is useful for sporadicity: if consecutive occurrences of e must be separated by at least θ , one may write $\Box(e \Rightarrow \widehat{\Box}_{<\theta} \neg e)$. The hatted form retains the triggering event as the time origin; ordinary bounded temporal operators do not express this particular reference shift directly. Related pointwise MTL constructions and the CASAAL translation are discussed in prior work [5]–[8].

2.2 Timed Büchi Automata

A TBA is a tuple $\mathcal{A} = (L, \ell_0, X, E, Acc, Inv)$. Here L is a finite set of locations, X a finite set of clocks, $E \subseteq L \times \text{Lab} \times \text{Guard} \times 2^X \times L$ is a set of action-labelled transitions with guards and reset sets, $Acc \subseteq L$ is the Büchi set, and Inv assigns a clock constraint to each location. A run on a timed word follows one transition per event, respects the invariant throughout each delay, tests guards at the event, and resets exactly the clocks on the transition. Acceptance requires infinitely many visits to Acc . For the translation theorem below, the initial clock values are existentially quantified. This matches the clock-consistent-extension semantics used by the Coq development; conventional zero initialization is a separate property. More explicitly, let v_i be the valuation tested at event i , and let $\Delta_i = t_{i+1} - t_i$. A run is an infinite sequence $(\ell_i, v_i)_{i \geq 0}$ with an initial nonnegative valuation v_0 . At event i , it chooses $(\ell_i, \alpha, g, Z, \ell_{i+1}) \in E$ such that the action satisfies α , the event valuation satisfies g , and the invariant of ℓ_i holds throughout the delay since the preceding event. The post-transition valuation is

$$v_{i+1}(x) = \begin{cases} \Delta_i, & x \in Z, \\ v_i(x) + \Delta_i, & x \notin Z, \end{cases}$$

and must satisfy $Inv(\ell_{i+1})$. For $i > 0$, the delay is $t_i - t_{i-1}$; the first event's pre-event valuation is existential. This event-indexed convention makes TBA clock values correspond directly to the clock values of an extended word.

3 CLOCKED-LTL TRANSLATION

3.1 Shared clock identities

The translation is structural. It maps Boolean and untimed temporal operators homomorphically and maps each primitive timed subformula to an LTL formula over an extended alphabet. The extended alphabet contains action atoms, clock comparisons $x \sim d$, a reset atom $\text{rst}(x)$, and a preservation atom $\text{unch}(x)$. These last atoms describe the clock update at the current event. Clock values are read before the event's reset, matching the guard semantics of a timed automaton.

For formula f , let $\text{Clock}(f)$ be its finite set of distinct primitive hatted timed subformulas. The clock associated with $F \in \text{Clock}(f)$ is $x = F$ itself. Thus two identical timed subformulas at different syntax-tree paths denote the same logical clock. Syntactically different subformulas remain distinct in this construction. The main result proves that the clock also serves all dynamic activations of that subformula. The clock count is therefore bounded by the number of distinct primitive timed subformulas, which is at most the number of timed occurrences in f . If structural subformulas are represented as shared nodes, the translation can share their LTL encodings as well; this does not remove the usual worst-case growth of an LTL-to-Büchi construction.

This sharing is semantic rather than an automaton-state minimization. The clock key is the normalized primitive formula, including its comparison and constant, so identical obligations have one clock identity before the propositional automaton is built. The reset trace is then chosen for that identity as a whole. In particular, the proof cannot justify the clock bound by assigning a fresh clock to each activation and subsequently identifying those clocks: it establishes directly that a single trace of reset decisions for the formula is sufficient.

3.2 Eight primitive clauses

Let F be a primitive hatted formula at path π , with operands p, q . Write $x = \text{clock_of}(F)$, $P = \mathcal{T}_{\pi 0}(p)$, $Q = \mathcal{T}_{\pi 1}(q)$, and define $\alpha W \beta = (\alpha U \beta) \vee G\alpha$. For upper comparisons $\lesseqgtr \in \{\leq, <\}$ and lower comparisons $\gtrless \in \{\geq, >\}$, complements are $\leq^c = >$, $<^c = \geq$, $\geq^c = <$, and $>^c = \leq$. The four clause shapes, each instantiated for its strict and non-strict bound, give all eight primitive cases:

$$\begin{aligned} \mathcal{T}_{\pi}(p \hat{U}_{\lesseqgtr} q) = & \mathbf{X}(((x \lesseqgtr d) \wedge \text{unch}(x) \wedge P) \\ & \mathbf{W}((x \lesseqgtr d) \wedge Q)), \end{aligned} \quad (5)$$

$$\begin{aligned} \mathcal{T}_{\pi}(p \hat{U}_{\gtrless} q) = & \text{rst}(x) \wedge \mathbf{X}((P \mathbf{U}((x \gtrless d) \wedge (P \mathbf{U} Q))) \\ & \vee (GP \wedge GFQ)), \end{aligned} \quad (6)$$

$$\begin{aligned} \mathcal{T}_{\pi}(p \hat{R}_{\lesseqgtr} q) = & \text{rst}(x) \wedge \mathbf{X}(Q \\ & \mathbf{W}((x \gtrless^c d) \vee (P \wedge Q))), \end{aligned} \quad (7)$$

$$\begin{aligned} \mathcal{T}_{\pi}(p \hat{R}_{\gtrless} q) = & \mathbf{X}(((x \gtrless^c d) \wedge \text{unch}(x)) \\ & \mathbf{W}((P \mathbf{R} Q) \vee ((x \gtrless^c d) \wedge P))). \end{aligned} \quad (8)$$

All clock atoms occur positively. Upper-bounded hatted Until uses weak Until because a clock-consistent trace must eventually cross its finite upper bound if no endpoint is

TABLE 1
Clock comparisons used by each hatted primitive.

Primitive role	Non-strict bound	Strict bound
$\hat{U}_{\leq d}$ endpoint	$x \leq d$	$x < d$
$\hat{U}_{\geq d}$ threshold	$x \geq d$	$x > d$
$\hat{R}_{\leq d}$ window end	$x > d$	$x \geq d$
$\hat{R}_{\geq d}$ pre-threshold	$x < d$	$x \leq d$

found. Lower-bounded Until retains its fairness branch, including GFQ , so that eventuality is not lost. The other weak-Until clause marks the end of a bounded Release window. Strict comparisons are represented directly; no strict bound is weakened by adjusting its constant.

The clauses have four distinct temporal roles. For upper-bounded Until, the clock is tested before any new reset: while its value remains in the allowed window, P must continue until Q appears; if the bound is passed first, the weak-until tail requires the continuation to remain valid. For lower-bounded Until, a finite path waits for the threshold and then starts the unbounded $P \mathbf{U} Q$ obligation. If the threshold is never reached, the second disjunct requires P at every position and Q infinitely often. Time divergence is what makes this infinite branch equivalent to a witness at distance at least d .

For upper-bounded Release, the initial Q is required and the weak-until stops either at the complementary clock comparison or when $P \wedge Q$ discharges the obligation. For lower-bounded Release, the clock remains unchanged before the threshold; after that point, the translated obligation is ordinary Release, with the endpoint alternative covering the transition into the threshold region. Thus Until clauses use existential witnesses, whereas Release clauses maintain a universal condition over the bounded region. The hatted shift is encoded by the outer $\mathbf{X}$ in every clause; for restart families, the separate $\text{rst}(x)$ atom denotes the reference timestamp at the current position.

3.3 Why hatted operators and shared clocks matter

The hatted operators preserve the triggering timestamp while moving logical evaluation to the next event. The same distinction appears in the clocked encoding: the initial $\mathbf{X}$ moves the temporal obligation forward, while the clock retains the reference selected by its reset trace.

Clock sharing has two layers. First, identical primitive formulas at different syntax-tree paths share one clock by construction. Second, that single clock must represent all pending activations. For example, in $\square(\text{req} \Rightarrow \diamond_{\leq 5} \text{ack})$, requests at times 0 and 2 can both be pending before an acknowledgement. Retaining the reference from time 0 suffices: an acknowledgement at time 4 is within five time units of both requests. Table 2 traces this case. Clock values are measured before the reset decision at each event; at the acknowledgement, the guard sees age four and the successful endpoint discharges both activations. For any newer request at time t_k between an older activation t_i and a witness t_j , $t_j - t_k \leq t_j - t_i$; the same endpoint therefore satisfies the newer upper-bound obligation. The argument depends on the required P condition holding

TABLE 2
One shared upper-Until clock for overlapping requests.

Event i	Action	Time t_i	Clock age before decision
0	<i>req</i>	0	0
1	<i>req</i>	2	2
2	<i>ack</i>	4	4

between activations and the endpoint, which is exactly what the Until residual records.

For the other primitive families, the canonical policy and residual predicates handle the direction in which restarting or preserving the clock is required (Supplement 2.6).

3.4 Canonical reset policy

The encoding quantifies over clock traces. To prove completeness with one clock for a formula that may be activated arbitrarily often, it is enough to construct one trace that gives priority to an older activation only while that activation remains useful. The proof records usefulness by residual obligations. For upper-bounded Until, the residual says that a witness is still reachable from the current clock age:

$$\begin{aligned} \text{URes}_{\preceq}(i, v, d, p, q) \iff \exists j \geq i. v + \text{elap}_w(i, j) \preceq d \\ \wedge w, j \models_{\text{MTL}} q \\ \wedge \forall k \in [i, j). \\ w, k \models_{\text{MTL}} p, \end{aligned}$$

where $\preceq$ is either $\leq$ or $<$. For lower-bounded Release, the residual records that every position beyond the threshold is still covered:

$$\begin{aligned} \text{RRes}_{\geq}(i, v, d, p, q) \iff \forall j \geq i. d \leq v + \text{elap}_w(i, j) \Rightarrow \\ \left(w, j \models_{\text{MTL}} q \right. \\ \left. \vee \exists k \in [i, j). \right. \\ \left. w, k \models_{\text{MTL}} p \right), \\ \text{RRes}_{>}(i, v, d, p, q) \iff \forall j \geq i. d < v + \text{elap}_w(i, j) \Rightarrow \\ \left(w, j \models_{\text{MTL}} q \right. \\ \left. \vee \exists k \in [i, j). \right. \\ \left. w, k \models_{\text{MTL}} p \right). \end{aligned}$$

The strict predicates preserve the strict comparison itself. They are defined and shifted in the Coq development (the full definitions and one-step lemmas are in Supplement 2.6).

The residual predicates are deliberately asymmetric. An upper-Until residual is existential: one still-reachable endpoint is enough, and that endpoint can discharge the obligation initiated at the current position. A lower-Release residual is universal: every position that has entered the threshold region must satisfy the Release condition. The canonical policy preserves a clock only when its residual remains true, so clock reuse follows from the semantic shape of the obligation rather than a syntactic rule such as “reset on every activation.” This distinction is needed for repeated activations, but the policy is not part of the generated automaton: it is a word-dependent witness used only to prove completeness.

TABLE 3
Canonical reset policy for each timed-operator family.

Primitive clock F	Reset decision at position i
$p \hat{U}_{\leq d} q$	Preserve iff $v \leq d$, $\text{URes}_{\leq}(i, v, d, p, q)$, and $w, i \not\models_{\text{MTL}} q$.
$p \hat{U}_{< d} q$	Preserve iff $v < d$, $\text{URes}_{<}(i, v, d, p, q)$, and $w, i \not\models_{\text{MTL}} q$.
$p \hat{U}_{\geq d} q, p \hat{U}_{> d} q, p \hat{R}_{\leq d} q,$ $p \hat{R}_{< d} q$ $p \hat{R}_{\geq d} q$	Reset iff $w, i \models_{\text{MTL}} F$.
$p \hat{R}_{> d} q$	Preserve iff $v < d$, $\text{RRes}_{\geq}(i, v, d, p, q)$, and $w, i \not\models_{\text{MTL}} p$.
	Preserve iff $v \leq d$, $\text{RRes}_{>}(i, v, d, p, q)$, and $w, i \not\models_{\text{MTL}} p$.

For clock $x = F$ with current value v , the canonical policy is summarized in Table 3. “Preserve iff” means that the clock is not reset exactly under the listed condition; in the remaining cases it is reset. For the restart families, the stated test is the reset condition. The canonical extension uses this decision for the formula-keyed clock and updates its value by

$$u_0 = 0, \quad u_{i+1} = \begin{cases} \Delta_i, & \text{if the policy resets at } i, \\ u_i + \Delta_i, & \text{otherwise.} \end{cases}$$

The policy depends on truth in the timed word and, through residuals, on its future. It is a semantic witness used in the completeness proof, not an algorithm executed by the translator. Its input is the timed formula itself, not a syntax-tree path, so identical occurrences share the same reset trace.

The two preservation cases retain an older reference only if the residual is still true and the current event has not discharged it. A later endpoint that satisfies the older upper-bounded Until also satisfies newer pending copies with shorter remaining prefixes. For lower-bounded Release, an old reference reaches the threshold no later than a newer one; the residual shift $(i, v) \mapsto (i + 1, v + \Delta_i)$ carries the remaining obligation. Conversely, lower-bounded Until and upper-bounded Release restart when the current formula is true; their translated continuation keeps any older obligations represented. The strict cases follow the same organization with strict residuals and guards.

4 FROM CLOCKED LTL TO A TBA

4.1 LTL-to-Büchi contract

The backend treats extended atoms as independent propositions. It returns a propositional Büchi automaton whose language is exactly that of the clocked-LTL formula. The theorem is stated for any such backend, deterministic or not.

Assumption 4.1 (Backend correctness). *For each formula F and each propositional word s over its atoms, $\text{PBA_accepts}(A, s)$ iff $s, 0 \models F$, where A is the backend automaton for F .*

This contract ranges over all propositional words, including assignments that do not correspond to any clock-consistent timed word. That stronger scope is used by relaxation.

4.2 Relaxation and reset completion

The clocked-LTL formula has positive occurrences of extended atoms. Its propositional language is consequently upward closed in those atoms. In each backend cube, relaxation deletes negative literals over extended atoms and irrelevant extended atoms; contradictory cubes are discarded. This preserves the accepted language because any satisfying assignment can be adjusted to the original cube while remaining a model of the positive formula.

Each clock belongs to one of two classes. A preservation clock is used with upper-bound tests and $\text{unch}(x)$, and a restart clock is used with lower-bound tests and $\text{rst}(x)$. Reset completion converts a relaxed cube into a TBA transition as follows:

- reset a preservation clock unless the cube contains $\text{unch}(x)$;
- reset a restart clock exactly when the cube contains $\text{rst}(x)$;
- remove reset and preservation markers, retaining clock tests as the transition guard and action literals as its label.

The asymmetry is essential. Additional resets only decrease preservation clock values and hence preserve upper tests. A restart clock can be reset only when explicitly required, since decreasing it could invalidate a lower test. The full monotonicity argument is given in Supplement 1.1.

Lemma 4.2 (Reset completion). *For every accepting run of the relaxed backend automaton on a clock-consistent extension, there is a clock-consistent extension with the same timed word and initial values whose resets are exactly those prescribed by reset completion, and the same run is accepting in the resulting TBA.*

Proposition 4.3 (Exactness of relaxation and reset completion). *Let f be well formed and let A satisfy Assumption 4.1 for $\mathcal{T}(f)$. The TBA obtained from A by relaxation and reset completion accepts exactly the timed words w for which $w, 0 \models_{\text{MTL}} f$, under existential initial clock values.*

The relaxation proof uses positivity of the clock atoms in $\mathcal{T}(f)$. A run of the original Büchi automaton remains a run after relaxation. In the reverse direction, take a run of the relaxed automaton and adjust each propositional assignment only on the extended atoms whose negative literals were deleted, making those atoms false; set irrelevant deleted atoms to true where needed to recover the original cube. The backend contract makes the adjusted word a model of $\mathcal{T}(f)$. On atoms that occur in $\mathcal{T}(f)$, the relaxed word only adds truth, so upward closure makes it a model too. Reset completion then assigns resets monotonically: preservation clocks become no larger, keeping upper guards true, and restart clocks become no smaller, keeping lower guards true. Thus the same Büchi run remains accepting. The proof is mechanized as `relax_sound`, `relax_complete`, and `complete_complete`.

Acceptance convention. The theorem quantifies existentially over the nonnegative clock values at the first event. This is a property of the accepted timed language; it does not by itself establish equivalence with an automaton whose clocks must start at zero. For example, a first transition

guarded by $x \geq 1$ may be enabled for one existential initial value and disabled when the first event is at time zero. The conventional initialization question is therefore outside Theorem 5.3; the Coq result is stated for the existential convention used throughout the translation proof.

5 CORRECTNESS AND CLOCK BOUND

An extended word ρ adds, at every position i , a value $\text{val}_\rho(i, x) \in \mathbb{R}_{\geq 0}$ and reset decision $\text{reset}_\rho(i, x)$ for each $x \in \text{Clock}(f)$. It is clock-consistent when, for $\Delta_i = t_{i+1} - t_i$,

$$\text{val}_\rho(i+1, x) = \begin{cases} \Delta_i, & \text{reset}_\rho(i, x), \\ \text{val}_\rho(i, x) + \Delta_i, & \text{otherwise.} \end{cases} \quad (9)$$

The value at position i is the value tested by guards before the reset decision at that position. Let $\text{same_base}(\rho, w)$ mean that w is the timed-word projection of ρ .

Theorem 5.1 (Correctness of the clock encoding). *For every well-formed formula f and timed word w ,*

$$w, 0 \models_{\text{MTL}} f \iff \exists \rho. \text{same_base}(\rho, w) \wedge \text{clock_consistent}(\rho) \wedge \rho, 0 \models_{\text{LTL}} \mathcal{T}(f).$$

The existential quantifier over ρ is taken jointly over all distinct timed subformulas. Soundness does not rely on choosing the canonical trace: each clock may follow any reset pattern allowed by the encoding, provided its value recurrence is respected. Completeness is the more delicate half because all activations that share a formula-keyed clock must be served simultaneously. The proof constructs the clock traces in parallel, using the residual predicates only for the preservation families and the direct restart rule for the remaining families. This result is the Coq theorem `EncodingCorrect_proved`. Its soundness direction holds for any clock-consistent reset trace. Completeness constructs one canonical trace for all clocks together. The detailed induction, including the strict-bound arithmetic, is in Supplement 2.

Corollary 5.2 (One clock per distinct primitive timed subformula). *All occurrences of the same primitive timed subformula use the same logical clock. The encoding uses exactly the clocks in $\text{Clock}(f)$, hence at most one clock per distinct primitive timed subformula, independently of the number of syntax-tree occurrences or run-time activations.*

Proof. Every clock in $\mathcal{T}(f)$ is indexed by an element of $\text{Clock}(f)$, and `T_uses_every_clock` proves that every such element occurs in $\mathcal{T}(f)$. A single extension assigns one value and reset trace to each element of $\text{Clock}(f)$. Theorem 5.1 proves that the canonical trace suffices for every pending activation of that formula. $\square$

Define `compile_with(A)` by applying relaxation and reset completion to backend automaton A . Timed acceptance is existential over clock-consistent extensions of the timed word and requires that reset decisions agree with the reset sets along an accepting run.

Theorem 5.3 (MTL-to-TBA correctness). *For every well-formed f , backend automaton A satisfying Assumption 4.1, and timed word w ,*

$$w, 0 \models_{\text{MTL}} f \iff \text{TBA_accepts}(\text{compile_with}(A), w).$$

In Coq this is `MTL_to_TBA_correct_with`; the backend correctness is an explicit hypothesis, not a project-specific axiom.

Proof. Combine Theorem 5.1 with Proposition 4.3. The forward direction uses the canonical extension and reset-completion lemma; the reverse direction uses upward closure under relaxation, the backend contract, and clock-encoding soundness. $\square$

5.1 Proof structure

The Coq proof separates the translation into local timed reasoning and structural composition. The local lemmas cover only the eight hatted constructors; Boolean, next, untimed Until and Release follow by induction. The ordinary timed operators have already been unfolded by the semantic identities, so they introduce no additional clock case. The full case-by-case proof and arithmetic lemmas are in Supplement 2.

Policy-independent soundness. Clock consistency supplies two inequalities. If x is not reset on $[i, j]$, then $\text{val}_\rho(j, x) = \text{val}_\rho(i, x) + \text{elap}_w(i, j)$, so its age at j is at least the physical time elapsed since i . If x was reset at i , then $\text{val}_\rho(j, x) \leq \text{elap}_w(i, j)$; any subsequent reset only decreases its age. Thus an upper guard at a preserved endpoint entails the matching upper bound on elapsed time, while a lower guard after a reset entails the matching lower bound. The inequalities preserve strictness: $x < d$ implies elapsed time $< d$, and $x > d$ implies elapsed time $> d$.

For upper-bounded hatted Until, the LTL Until provides an endpoint, its clock guard bounds the distance, and the translated left operand supplies the open-interval prefix. Lower-bounded hatted Until has a finite branch with a threshold guard and an infinite branch $GP \wedge GFQ$. The finite branch gives the lower bound directly; in the infinite branch, divergence supplies a sufficiently late q witness. For strict lower bounds, strict monotonicity between positions i and $i + 1$ upgrades the distance from the $i + 1$ tail to a strictly greater distance from the original reference. The two Release families use weak Until to express a bounded window: upper-bounded Release ends when the complementary upper test becomes true or when $p \wedge q$ discharges the obligation; lower-bounded Release preserves its clock before the threshold until either p discharges it or ordinary Release takes over. These arguments use no property of the reset policy beyond clock consistency.

Completeness with one trace per formula. Completeness uses the canonical extension from Definition 2.1; the extension is shared by all occurrences and activations of each formula-keyed clock. For upper-bounded Until, the induction maintains, up to a selected endpoint j ,

$$\text{val}_{\rho^*}(n, x) + \text{elap}_w(n, j) \lesssim d.$$

If an older reference remains useful, its residual ensures this inequality continues to hold and also covers newer activations. Otherwise the policy resets and starts the current

TABLE 4

Coq lemmas for the primitive timed proof cases. Each row denotes both strict and non-strict comparisons.

Primitive family	Soundness	Canonical completeness
$\hat{U}_{\leq d}$	<code>sound_UhatLe</code>	<code>complete_UhatLe</code>
$\hat{U}_{\geq d}$	<code>sound_UhatGe</code>	<code>complete_UhatGe</code>
$\hat{R}_{\leq d}$	<code>sound_RhatLe</code>	<code>complete_RhatLe</code>
$\hat{R}_{\geq d}$	<code>sound_RhatGe</code>	<code>complete_RhatGe</code>
$\hat{U}_{< d}$	<code>sound_UhatLt</code>	<code>complete_UhatLt</code>
$\hat{U}_{> d}$	<code>sound_UhatGt</code>	<code>complete_UhatGt</code>
$\hat{R}_{< d}$	<code>sound_RhatLt</code>	<code>complete_RhatLt</code>
$\hat{R}_{> d}$	<code>sound_RhatGt</code>	<code>complete_RhatGt</code>

TABLE 5

Scope of the mechanized translation theorem.

Component	Status
Ordinary-operator definitions and point-wise compatibility	Coq-verified
Eight hatted clocked-LTL clauses	Coq-verified
Canonical reset policy and residual completeness	Coq-verified
Relaxation and reset completion	Coq-verified
LTL-to-Büchi language equivalence	Explicit assumption
Parser, backend process, output printer	Outside theorem

obligation with the required remaining budget. The first suitable q endpoint closes the weak-until requirement. The strict case uses the strict invariant with $< d$.

For lower-bounded Until, each activation of a true formula resets its clock. If a threshold is reached, the finite branch of the translated formula continues with $P \cup Q$. If it is never reached, the active-obligation invariant implies that p persists and q recurs, yielding the fairness branch. For upper-bounded Release, restarting at a true activation lets the weak-until clause track the current bounded window until $p \wedge q$ or the complementary clock test ends it. For lower-bounded Release, the invariant is the corresponding residual predicate; while the policy preserves the clock, its value advances by Δ_i , and the residual shifts from (i, v) to $(i + 1, v + \Delta_i)$. That shift retains the coverage of every position beyond the threshold. These cases establish the translated formula at each syntactic occurrence by induction, while the clock key identifies duplicate timed subformulas.

The local cases are assembled into the theorem `EncodingCorrect_proved`. Its completeness argument proves equivalence for every node identified by a syntax-tree path, not only for the root. This stronger induction invariant is needed because repeated formula occurrences share the same clock while the induction traverses separate paths.

What the mechanization assumes. The proof boundary is summarized in Table 5. In particular, the correctness of the untimed backend is an explicit proposition-level assumption; the timed proof does not depend on Spot or on determinism. In the included development, the theorem `MTL_to_TBA_correct_with` accepts that contract as an argument. The parser, backend invocation and output printer are not part of this theorem.

6 RELATED WORK

MTL and its dense-time pointwise semantics have a substantial literature, including decidability and expressiveness results [1], [2], [4]–[6]. The one-sided-bounded fragment is a useful compromise: full dense-time MTL does not in general admit finite timed-automaton translations, while $\text{MTL}_{0,\infty}$ retains common response and separation requirements [2], [3]. The hatted operators in this paper make the distinction between a timing reference and the current logical position explicit. They are primitive constructors in the logic and proof, so their strict pointwise behavior is not delegated to an unverified preprocessing pass.

Translations of timed logics. The closest complete automata construction for $\text{MTL}_{0,\infty}$ is CASAAL, which uses a dedicated tableau-style method and derives deterministic approximations when controller synthesis requires them [7], [8]. Bulychev et al. study controller synthesis for the same fragment [3]. Our objective is complementary: the formula is compiled to an exact nondeterministic TBA, which can then be combined with a system model. For MITL, classical constructions use temporal testers, networks of automata, or one-clock alternating automata [9]–[11]. Tools such as MightyL and MITL2Timed provide practical translations and UPPAAL-oriented workflows [12], [13]; they target a different logic and architecture. Existing techniques for merging timed obligations are related to our clock-reuse problem, but the residual predicates here are tailored to one-sided bounds and are part of the machine-checked completeness proof.

Clocked temporal logic and subformula clocks. CLTL_{oc} extends LTL with clock comparisons and reset predicates and reduces satisfiability to a theory solver; it has also been used to encode metric logics [14]. Our clocked-LTL formula uses a similar propositional vocabulary but serves as an intermediate representation for automaton synthesis. The backend treats clock tests and reset markers as propositions; positive clock atoms make the language upward closed, while preservation markers and reset completion recover their timed meaning. This relationship motivates the explicit relaxation proof rather than assuming that the backend understands clocks.

One clock per subformula is also familiar from event-clock logics and subformula-based translations [15]–[17]. Those clocks typically record time since, or until, an event. Here a clock is keyed by a timed formula and may face multiple overlapping activations of that same formula. The canonical trace resolves those competing references; its correctness is proved for each strict and non-strict primitive.

Mechanized translations and trusted backends. Mechanized work has verified LTL-to-Büchi translations and model checkers [18]–[21]. For metric logics, PVS developments formalize corrected decision procedures for MITL [22]; Coq has also been used for quantitative MTL monitoring [23]. These are complementary targets: our theorem concerns infinite dense-time words and the formula-keyed clocks required by an MTL-to-TBA construction. The theorem deliberately assumes the correctness of the LTL-to-Büchi backend. A verified LTL translator could discharge that contract if its output were adapted to the automaton interface used by the Coq development. Our result thus

isolates the timed translation proof from the backend rather than claiming that Spot itself is verified.

7 CONCLUSION

We have presented a mechanically verified translation from $\text{MTL}_{0,\infty}$ to timed Büchi automata. Hatted operators make the timing reference explicit; the clocked-LTL clauses cover strict and non-strict bounds; and relaxation with reset completion turns a correct propositional Büchi backend into a timed automaton. The canonical reset policy and residual obligations establish that one clock per distinct primitive timed subformula suffices even with arbitrarily many overlapping activations. Coq proves both semantic preservation and the end-to-end result `MTL_to_TBA_correct_with` under the stated backend contract.

REFERENCES

- [1] R. Koymans, “Specifying real-time properties with metric temporal logic,” *Real Time Syst.*, vol. 2, no. 4, pp. 255–299, 1990. [Online]. Available: <https://doi.org/10.1007/BF01995674>
- [2] R. Alur, T. Feder, and T. A. Henzinger, “The benefits of relaxing punctuality,” *Journal of the ACM (JACM)*, vol. 43, no. 1, pp. 116–146, 1996.
- [3] P. Bulychev, A. David, K. Larsen, and G. Li, “Efficient controller synthesis for a fragment of $\text{MTL}(0,\infty)$,” *Acta Informatica*, vol. 51, no. 3–4, pp. 165–192, Jan. 2014.
- [4] D. D’Souza and P. Prabhakar, “On the expressiveness of MTL in the pointwise and continuous semantics,” *International Journal on Software Tools for Technology Transfer*, vol. 9, no. 1, pp. 1–4, 2007.
- [5] J. Ouaknine and J. Worrell, “On the decidability of metric temporal logic,” in *20th Annual IEEE Symposium on Logic in Computer Science (LICS 2005)*. IEEE Computer Society, 2005, pp. 188–197.
- [6] P. Bouyer, F. Chevalier, and N. Markey, “On the expressiveness of TPTL and MTL,” in *Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2005)*, ser. Lecture Notes in Computer Science, vol. 3821. Springer, 2005, pp. 432–443.
- [7] G. Li, P. G. Jensen, K. G. Larsen, A. Legay, and D. B. Poulsen, “Practical controller synthesis for $\text{MTL}_{0,\infty}$,” in *Proceedings of the 24th ACM SIGSOFT International SPIN Symposium on Model Checking of Software*, ser. SPIN 2017. New York, NY, USA: Association for Computing Machinery, 2017, pp. 102–111. [Online]. Available: <https://doi.org/10.1145/3092282.3092303>
- [8] G. Li, “CASAAL: A tool for translating $\text{MTL}_{0,\infty}$ into timed Büchi automata,” Online, 2017, available at the CASAAL tool page. [Online]. Available: <https://lcs.ios.ac.cn/~ligy/tools/CASAAL/>
- [9] O. Maler, D. Nickovic, and A. Pnueli, “From MITL to timed automata,” in *Formal Modeling and Analysis of Timed Systems (FORMATS 2006)*, ser. Lecture Notes in Computer Science, vol. 4202. Springer, 2006.
- [10] T. Ferrère, O. Maler, D. Ničković, and A. Pnueli, “From real-time logic to timed automata,” *Journal of the ACM*, vol. 66, no. 3, 2019, art. no. 19.
- [11] T. Brihaye, M. Estiévenart, and G. Geeraerts, “On MITL and alternating timed automata over infinite words,” 2014. [Online]. Available: <https://arxiv.org/abs/1406.4395>
- [12] T. Brihaye, G. Geeraerts, H.-M. Ho, and B. Monmege, “MightyL: A compositional translation from MITL to timed automata,” in *Computer Aided Verification*, R. Majumdar and V. Kunčák, Eds. Cham: Springer International Publishing, 2017, pp. 421–440.
- [13] “MITL2Timed: Modifying Oddoux and Gastin LTL2BA to timed automata generation,” GitHub repository. [Online]. Available: <https://github.com/yzh89/MITL2Timed>
- [14] M. M. Bersani, M. Rossi, and P. San Pietro, “A tool for deciding the satisfiability of continuous-time metric temporal logic,” *Acta Informatica*, vol. 53, 2016.
- [15] R. Alur, L. Fix, and T. A. Henzinger, “Event-clock automata: A determinizable class of timed automata,” *Theoretical Computer Science*, vol. 211, no. 1–2, pp. 253–273, 1999.
- [16] J.-F. Raskin and P.-Y. Schobbens, “The logic of event clocks: Decidability, complexity and expressiveness,” *Journal of Automata, Languages and Combinatorics*, vol. 4, no. 3, pp. 247–286, 1999.

- [17] D. Ničković and N. Piterman, “From MTL to deterministic timed automata,” in *Formal Modeling and Analysis of Timed Systems (FORMATS 2010)*, ser. Lecture Notes in Computer Science, vol. 6246. Springer, 2010, pp. 152–167.
- [18] A. Schimpf, S. Merz, and J.-G. Smaus, “Construction of Büchi automata for LTL model checking verified in Isabelle/HOL,” in *Theorem Proving in Higher Order Logics (TPHOLs 2009)*, ser. Lecture Notes in Computer Science, vol. 5674. Springer, 2009, pp. 424–439.
- [19] J. Esparza, P. Lammich, R. Neumann, T. Nipkow, A. Schimpf, and J.-G. Smaus, “A fully verified executable LTL model checker,” in *Computer Aided Verification (CAV 2013)*, ser. Lecture Notes in Computer Science, vol. 8044. Springer, 2013, pp. 463–478.
- [20] A. Schimpf and P. Lammich, “Converting linear-time temporal logic to generalized Büchi automata,” *Archive of Formal Proofs*, 2014, https://isa-afp.org/entries/LTL_to_GBA.html, formal proof development.
- [21] B. Seidl and S. Sickert, “A compositional and unified translation of LTL into ω -automata,” *Archive of Formal Proofs*, https://isa-afp.org/entries/LTL_Master_Theorem.html, 2019.
- [22] N. Roohi and M. Viswanathan, “Revisiting MITL to fix decision procedures,” in *Verification, Model Checking, and Abstract Interpretation (VMCAI 2018)*, ser. Lecture Notes in Computer Science, vol. 10747. Springer, 2018.
- [23] A. Chattopadhyay and K. Mamouras, “A verified online monitor for metric temporal logic with quantitative semantics,” in *Runtime Verification (RV 2020)*, ser. Lecture Notes in Computer Science, vol. 12399. Springer, 2020.